Assignment - 5.

1) 8-bit encoders, $BR = 96 \times 10^6$ bps.

$BR = n \cdot f_s \rightarrow f_s = \frac{BR}{n} = \frac{96 \times 10^6}{8} = 12 \times 10^6$ samp/sec

nyquist $\Rightarrow f_s \geq 2B$.

$\Rightarrow 12 \times 10^6 \geq 2B \Rightarrow B \leq 6 \times 10^6$ Hz.

$\Rightarrow B_{max} = 6 \times 10^6$ Hz.

For sinusoidal input :- $SQNR\,(dB) = 6.02n + 1.76$.

$\Rightarrow 6.02(8) + 1.76 = 49.92$ dB.

$A = \pm 4V \Rightarrow$ range $= 8V$

no. of quantization levels $= 2^8 = 256$.

Quantization step $= \frac{\text{range}}{\text{levels}} = \frac{8}{2^8} = \frac{1}{32} = 0.03125$.

Quantization error $= \frac{\text{step size}}{2} = \frac{0.03125}{2} = 0.015625$.

2) $T = 15s$, $f_s = 10 \times 10^3$ Hz.

$SQNR = 48$ dB $= 6.02n + 1.76$, $n = \frac{48 - 1.76}{6.02} = 7.681$

$\Rightarrow n = 8$ bits/sample.

no. of samples $= 15 \times 10 \times 10^3 = 15 \times 10^4$; tot no. of bits $= 15 \times 10^4 \times 8$.

$= 120 \times 10^4 = 1.2$ Mb

3) $s = \pm\sqrt{E_b}$, $E_b = 2$ $r = s + n$ $n \sim N(0, 0.5)$

ML rule $\Rightarrow$ $r > 0 \Rightarrow s = +\sqrt{E_b}$

$r < 0 \Rightarrow s = -\sqrt{E_b}$.

$\Rightarrow$ Symbol detected $= \begin{cases} +\sqrt{2} & r > 0 \\ -\sqrt{2} & r < 0. \end{cases}$ prob. of error given $s = +\sqrt{2}$.

$\Rightarrow$ error occurs when $r < 0 \rightarrow \sqrt{2} + n < 0$.

$P_e = Pr(r < 0 \mid s = +\sqrt{2}) = Pr(\sqrt{2} + n < 0) = Pr(n < -\sqrt{2})$

$n \sim N(0, 0.5) \Rightarrow \sigma = \sqrt{0.5}$. $Pr(n < -\sqrt{2}) = Pr(z < \frac{-\sqrt{2}}{\sqrt{0.5}})$

$\Rightarrow Pr(z < -2) = Q(2)$

4) $s = \pm\sqrt{E_b}$ $E_b = 4$ $r = s + n$ $n \sim N(0, 1)$

$\gamma = 0.5$ detected $\Rightarrow \begin{cases} +\sqrt{E_b}, & r > \gamma \\ -\sqrt{E_b}, & r < \gamma. \end{cases} \Rightarrow \begin{cases} +2, & r > 0.5 \\ -2 & r < 0.5 \end{cases}$

$P_e = Pr(r < 0.5 \mid s = +2) = Pr(2 + n < 0.5) = Pr(n < -1.5)$

$P_e = Q(1.5)$

$P_e = Pr(r > 0.5 / s = -2) = Pr(-2 + n > 0.5) = Pr(n > 2.5)$

$P_e = Q(2.5)$; $P_e = \frac{1}{2}(Q(1.5) + Q(2.5))$

Optimal detector $\Rightarrow P_e = Q(2)$

$(P_e)_{biased} > (P_e)_{optimal}$

5). $I, Q \in \{\pm 1\}$, $\sigma_I^2 = 1$, $\sigma_Q^2 = 4$.

$r_I = 0.2$, $r_Q = -0.5$

$P_e = Pr(r_I < 0 | I = +1) + Pr(r_I > 0 | I = -1)$

$= \frac{1}{2}[Pr(1 + n < 0) + Pr(-1 + n > 0)] = Pr(n < -1) = Q(1)$

$P_e = \frac{1}{2}(Pr(r_Q < 0 | Q = +1) + Pr(r_Q > 0 | Q = -1)]$.

$= \frac{1}{2}[Pr(1 + n < 0) + Pr(-1 + n > 0)] = Pr(n < -1)$

$\Rightarrow Pr(Z < \frac{1}{2}) = Q(0.5)$

$(P_e)_I = Q(1)$ $(P_e)_Q = Q(0.5)$.

Symbol transmitted $\Rightarrow I = +1, Q = -1$.

dominant error source $= (P_e)_Q$.

6) 8 - PSK $\frac{E_b}{N_0} = 10dB$. $E_s = kE_b$, $k = \log_2 M$.

$\frac{E_s}{N_0} = \frac{3E_b}{N_0} = 10\log 3 + \left(\frac{E_b}{N_0}\right) dB = 4.771 + 10 = 14.771 \, dB$. $E_s = 3E_b$ $k = 3$.

$(P_e)_{sym} = 2Q\left(\sqrt{\frac{2E_s}{N_0}} \sin\left(\frac{\pi}{8}\right)\right) = 2Q(\sqrt{60} \times 0.3827)$

$= 2Q(2.9642)$ $(P_e)_{bit} = \frac{(P_e)_{sym}}{\log_2 M} = \frac{2Q(2.96)}{3}$.

7. Constellation point of QAM.

$= (\pm a, 0), (0, \pm a), \left(\pm \frac{b}{\sqrt{2}}, \pm \frac{b}{\sqrt{2}}\right)$

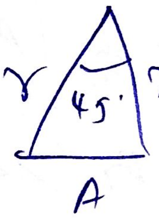
$\left(a - \frac{b}{\sqrt{2}}\right)^2 + \left(0 - \frac{b}{\sqrt{2}}\right)^2 = A^2$

$a^2 + \frac{b^2}{2} - \sqrt{2}ab + \frac{b^2}{2} = A^2 \Rightarrow (a^2+b^2) - \sqrt{2}ab = A^2$

optimal QAM (minimum energy).

$\Rightarrow b = \sqrt{2}a \Rightarrow (a^2 + \cancel{2a^2}) \cancel{= 2a^2} = A^2.$

$A = a \Rightarrow b = \sqrt{2}A.$



$\cos 45 = \frac{r^2 + r^2 - A^2}{2r^2} = \frac{2r^2 - A^2}{2r^2} = \frac{1}{\sqrt{2}}.$

$\Rightarrow 2r^2 - A^2 = \sqrt{2}r^2 \Rightarrow (2-\sqrt{2})r^2 = A^2.$

$r = \frac{A}{\sqrt{2-\sqrt{2}}}$, for 8-QAM, E_{sym} = Sq magni.

→ for 4 points of radi of circle $E_s = a^2$.

for 4 pts on radi of b $E_s = b^2$

$(E_{avg})_{QAM} = \frac{1}{8}(4a^2 + 4b^2) = \frac{1}{2}(a^2+b^2) = \frac{1}{2}(A^2 + 2A^2)$

$\Rightarrow \frac{3}{2}A^2 = 1.5A^2.$

for 8-PSK $\Rightarrow E_{avg} = \frac{1}{8}(8r^2) = r^2.$

$\Rightarrow \frac{A^2}{2-\sqrt{2}} \geq 1.5A^2 = 1.707A^2$.

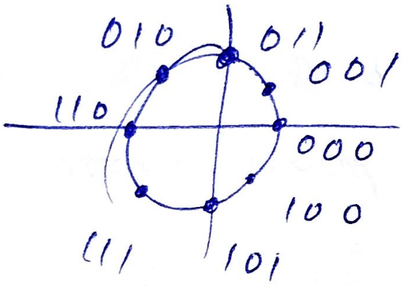
$\Rightarrow (E_{avg})_{PSK} > (E_{avg})_{QAM}$. Relative power adv

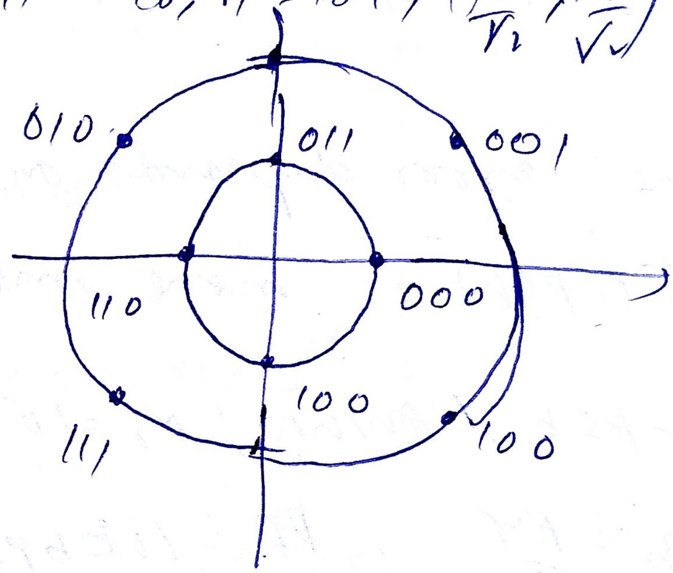
$\Rightarrow \frac{1.707A^2}{1.5A^2} = 1.38$.

$10\log(1.38) = 0.5616\,dB$.

$(r,0) = 000$ $(0,r) = 011$ $\left(\frac{r}{\sqrt{2}}, \frac{r}{\sqrt{2}}\right) = 001$, $\left(\frac{-r}{\sqrt{2}}, \frac{r}{\sqrt{2}}\right) = 010$

$(-r,0) = 110$ $\left(\frac{-r}{\sqrt{2}}, \frac{-r}{\sqrt{2}}\right) = 111$ $(0,r) = 101$, $\left(\frac{r}{\sqrt{2}}, \frac{-r}{\sqrt{2}}\right) = 100$.





(bit rate) $B_r = 90\,Mbps$.

$SR = \frac{BR}{\log_2 M} = \frac{90}{3b} = 30\,Mbps$.

$P_e \propto Q\left(\frac{d_{min}}{2\sigma}\right) = \text{same} \Rightarrow d_{min} \propto \sigma \Rightarrow d_{min} \propto E_s$.

(8-PSK) $d_{min} = 2r\sin\left(\frac{45}{2}\right)$.

$\frac{d_{min}^2}{E_s} = \frac{A^2}{\frac{3}{2}A^2} = \frac{2}{3} = 0.34$.

$$\left(\frac{d_{min}^2}{E_s}\right)_{PSK} > \left(\frac{d_{min}^2}{E_s}\right)_{QAM.}$$

$(SNR)_{PSK} > (SNR)_{QAM}$. $\Rightarrow$ For 8PSK $\Rightarrow s = re^{j\phi}$.

after phase error $= re^{j(\phi+\theta)}$.

phase changes, magnitude constant.

For 8-QAM $\Rightarrow s = I + jQ$.
$r_I = I\cos\theta - Q\sin\theta$
$r_Q = I\sin\theta - Q\cos\theta$

here error depends on both I & Q.

$\Rightarrow$ (8-PSK) is more immune to phase error

8) b-PSK, AWGN, $N_0 = 10^{-10}$ W/Hz.

$E_b = \frac{A^2T}{2}$, DR = 10kbps, $Pe = 10^{-6}$.

$$Pe = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = Q\left(\sqrt{\frac{2A^2T/2}{10^{-10}}}\right) \Rightarrow 10^{-6} = Q\left(\sqrt{\frac{A^2T}{10^{-10}}}\right)$$

$$\Rightarrow \sqrt{\frac{A^2T}{10^{-10}}} \simeq 4.75 \Rightarrow A^2T = (4.75)^2 \times 10^{-10}.$$

$$\Rightarrow A^2T = 22.56 \times 10^{-10}$$

DR = 10 kbps $\Rightarrow 10 \times 10^3 \Rightarrow T = 10^{-4}$.

$A^2 = 22.56 \times 10^{-10+4} = 2.25 \times 10^{-5}$

$A = 4.75 \times 10^{-3} = 4.75\ mV$

$DR = 100\ kbps = 10^5 \Rightarrow T = 10^{-5}$.

$\Rightarrow A^2 = 22.56 \times 10^{-10+5} = 22.56 \times 10^{-5}$

$A = 1.5 \times 10^{-2} = 15\ mV$.

$DR = 1\ Mbps \Rightarrow 10^6 \quad T = 10^{-6}$.

$A^2 = 22.56 \times 10^{-10+6} = 22.56 \times 10^{-4} \quad A = 4.75 \times 10^{-2}$

$\Rightarrow 47.5\ mV$.

5) $SR = 2400$ sym/second.

$P_e = 10^{-5} \quad BR = 4800$ bps.

bits per symbol $= \frac{4800}{2400} = 2. \quad M = 2^2 = 4 \Rightarrow 4\text{-QAM}$.

$P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = 10^{-5} \Rightarrow \sqrt{\frac{2E_b}{N_0}} = 4.265$.

$\frac{E_b}{N_0} = \frac{(4.265)^2}{2} = 9.1 \quad 10 \log 9.1 = 9.6\ dB$.

$BR = 9600$ bps $\Rightarrow$ no. of bits $= \frac{9600}{2400} = 4$.

$M = 2^4 = 16 \Rightarrow 16\text{-QAM} \Rightarrow P_e = \frac{4}{k}\left(1 - \frac{1}{\sqrt{M}}\right) Q\left(\sqrt{\frac{3kE_b}{(M-1)N_0}}\right)$

$\Rightarrow \frac{4}{4}\left(1 - \frac{1}{\sqrt{16}}\right) Q\left(\sqrt{\frac{3 \times 4 \times E_b}{15 \times N_0}}\right) \Rightarrow \left(1 - \frac{1}{4}\right) Q\left(\sqrt{\frac{4E_b}{5N_0}}\right)$

$$\Rightarrow \frac{3}{4} Q\left(\sqrt{\frac{4E_b}{5N_0}}\right) = 10^{-5} \Rightarrow Q\left(\sqrt{\frac{4E_b}{5N_0}}\right) = \frac{4}{3}\times 10^{-5}$$

$$\Rightarrow \sqrt{\frac{4E_b}{5N_0}} = 4.2.$$

$$\frac{E_b}{N_0} = (4.2)^2 \times \frac{5}{4} = 22.05$$

$10\log(22.05) = 13.4\,dB$ $\quad$ $BR = 19200\,bps \Rightarrow k = \frac{19200}{2400} = 8.$

$\Rightarrow M = 2^8 = 256.$

$$256\text{-QAM} \Rightarrow P_e = \frac{4}{8}\left(1-\frac{1}{\sqrt{256}}\right) Q\left(\sqrt{\frac{3\times 8E_b}{255N_0}}\right).$$

$$= \frac{1}{2}\left(1-\frac{1}{15}\right) Q\left(\sqrt{\frac{8E_b}{85N_0}}\right) = \frac{1}{2}\times\frac{14}{15} Q\left(\sqrt{\frac{8E_b}{85N_0}}\right) = \frac{7}{15} Q\left(\sqrt{\frac{8E_b}{85N_0}}\right)$$

$$\Rightarrow Q\left(\sqrt{\frac{8E_b}{85N_0}}\right) = \frac{15}{7}\times 10^{-5}$$

$$\Rightarrow \sqrt{\frac{8E_b}{85N_0}} = 4.1 \Rightarrow \frac{E_b}{N_0} = \frac{85}{8}(4.1)^2 \Rightarrow \frac{E_b}{N_0} = 178.6.$$

$10\log(178.6) = 22.5\,dB.$ $\Rightarrow$ higher DR = higher $\frac{E_b}{N_0}$.

10) $\sqrt{2}\, r_1 = d \Rightarrow r_1 = \frac{d}{\sqrt{2}}$, $\quad \frac{r_1^2 + r_2^2 - d^2}{2r_2^2} = \frac{1}{\sqrt{2}}$

$d = r_2\left(\sqrt{2-\sqrt{2}}\right).$ $\quad$ $r_2 = \frac{d}{\sqrt{2-\sqrt{2}}}.$ $\quad$ $P_e \propto Q\left(\frac{d_{min}^2}{\sigma}\right).$

Compare using $\frac{d_{min}^2}{E_s}$; for 4-PSK

$$\frac{d_{min}^2}{E_s} = \frac{d^2}{r_1^2}$$

for 8-PSK $\frac{d_{min}^2}{E_s} = \frac{d^2}{r_2^2}$

$$\Rightarrow \frac{d^2}{(d/\sqrt{2})^2} = 2 \qquad \Rightarrow \frac{d^2}{(2/\sqrt{2-\sqrt{2}})^2}$$

$$\frac{E_{s2}}{E_{s1}} = \frac{2}{2-\sqrt{2}} = \frac{1}{1-\frac{1}{\sqrt{2}}} = 3.414$$

$$\frac{P_{8-PSK}}{P_{4-PSK}} = 3.414 \approx 5.33 \text{ dB}$$

11) Decision boundaries are along the midpoints of adjacent points & along $\pm 2, \pm 4, 0$ in X & Y dim.

12) $\{\phi_{nl}(t), n = 1, 2, \dots N\}$; $\{s_{ml}(t), m = 1, 2, \dots N\}$

$\phi_n(t) = \sqrt{2}\, R\{\phi_{nl}(t)\, e^{j2\pi f_0 t}\}$ $\tilde{\phi}_n(t) = -\sqrt{2}\, I\{\phi_{nl}(t)\, e^{j2\pi f_0 t}\}$

$\{\phi_n(t), \tilde{\phi}_n(t), n = 1, 2, \dots N\}$ 2N orthonormal basis.

$s_m(t) = R\{s_{ml}(t)\, e^{j2\pi f_0 t}\}$, $m = 1, 2, \dots M$.

$$\phi_n(t) = \sqrt{2} \cdot \frac{1}{2} [\phi_{nl}(t)\, e^{j2\pi f_0 t} + \phi_{nl}^*(t)\, e^{-j2\pi f_0 t})$$

$$= \frac{1}{\sqrt{2}} [\phi_{nl}(t)\, e^{j2\pi f_0 t} + \phi_{nl}^*(t)\, e^{-j2\pi f_0 t}]$$

$$\tilde{\phi}_n(t) = -\sqrt{2}\,\frac{1}{2j}\left[\phi_{nl}(t)\,e^{j2\pi f_0 t} - \phi_n^*(t)\,e^{-j2\pi f_0 t}\right].$$

$$= \frac{j}{\sqrt{2}}\left[\phi_{nl}(t)\,e^{j2\pi f_0 t} - \phi_n^*(t)\,e^{-j2\pi f_0 t}\right].$$

orthonormality, $\langle \phi_n, \phi_m \rangle = \delta_{nm}$.

$$\Rightarrow \int \phi_{nl}(t)\,\phi_{ml}^*(t)\,dt = \delta_{nm}.$$

$$\Rightarrow \int \frac{1}{\sqrt{2}}\left[\phi_{nl}(t)\,e^{j2\pi f_0 t} + \phi_{nl}^*(t)\,e^{-j2\pi f_0 t}\right] \times$$

$$\frac{1}{\sqrt{2}}\left[\phi_{nl}(t)\,e^{j2\pi f_0 t} + \phi_{ml}^*(t)\,e^{-j2\pi f_0 t}\right] dt.$$

$$= \frac{1}{2}\int \left[\phi_{nl}(t)\,\phi_{ml}(t)\,e^{j4\pi f_0 t} + \phi_{nl}^*(t)\,\phi_{ml}^*(t)\,e^{-j4\pi f_0 t}\right] dt$$

$$= \int \phi_{nl}(t)\,\phi_{ml}^*(t)\,e^{j4\pi f_0 t}\,dt = \delta_{nm}.$$

Similarly, $\langle \tilde{\phi}_n, \tilde{\phi}_m \rangle = \delta_{nm}$.

$$\langle \phi_n, \tilde{\phi}_m \rangle = \int \frac{1}{\sqrt{2}}\left(\phi_{nl}(t)\,e^{j2\pi f_0 t} + \phi_{nl}^*(t)\,e^{-j2\pi f_0 t}\right) \times$$

$$\frac{j}{\sqrt{2}}\left[\phi_{nl}(t)\,e^{j2\pi f_0 t} - \phi_{nl}^*(t)\,e^{-j2\pi f_0 t}\right] dt.$$

$$\Rightarrow \frac{j}{2}\cdot\left[\phi_{nl}(t)\,e^{j2\pi f_0 t}\right]^2 - \left[\phi_{nl}^*(t)\,e^{-j2\pi f_0 t}\right]^2 dt.$$

$= 0 \Rightarrow$ orthonormal basis.

$\Rightarrow \phi_{nl}$ & $\hat{\phi}_{nl}$ $\Rightarrow$ 2N dimension

$$s_{ml}(t) = \sum_{n=1}^{N} a_{mn} \phi_{nl}(t)$$

$$\Rightarrow s_m(t) = \mathcal{R}\{\sum a_{mn} \phi_{nl}(t) e^{j2\pi f_0 t}\}$$

$$= \sum \mathcal{R}\{a_{mn} \phi_{nl}(t) e^{j2\pi f_0 t}\} \quad a_{mn} = \alpha + j\beta$$

$$\Rightarrow s_m(t) = \sum \alpha \phi_n(t) + \beta \tilde{\phi}_n(t) \quad \Rightarrow \text{bandpass signal}$$

13). $s_1, s_2, s_3, s_4 \rightarrow$ 0 to 4 seconds.

basis functions.

$$\phi_1(t) = \begin{cases} 1 & 0 \le t < 1 \\ 0 & \text{else} \end{cases} \quad \phi_2(t) = \begin{cases} 1 & 1 \le t < 2 \\ 0 & \text{else} \end{cases}$$

$$\phi_3(t) = \begin{cases} 1 & 0 \le t < 3 \\ 0 & \text{else} \end{cases} \quad \phi_4(t) = \begin{cases} 1 & 3 \le t < 4 \\ 0 & \text{else} \end{cases}$$

4-indepen. intervals. $\Rightarrow$ dimension = 4.

$$s_1(t) = 2\phi_1(t) - \phi_2(t) - \phi_3(t) - \phi_4(t)$$

$$s_2(t) = \phi_1(t) - \phi_2(t) + \phi_3(t) - \phi_4(t)$$

$$s_3(t) = -2\phi_1(t) + \phi_2(t) + \phi_3(t)$$

$$s_4(t) = \phi_1(t) - 2\phi_2(t) - 2\phi_3 + 2\phi_4(t)$$

$\Rightarrow s_1 = [2, -1, -1, -1]$, $s_2 = [1, -1, 1, -1]$

$s_3 = [-2, 1, 1, 0]$, $s_4 = [1, -2, -2, 2]$

min dist $\Rightarrow S_1S_3 = \sqrt{1^2+2^2} = \sqrt{3} \Rightarrow d_{min}$

$S_2S_3 = \sqrt{3^2+2^2+1^2} = \sqrt{14}$

$S_1S_2 = \sqrt{4^2+2^2+2^2+1} = 5$ $\quad S_1S_4 = \sqrt{1+1+1+3^2} = \sqrt{12}$

(4) $N_0/2 = 1 \Rightarrow N_0 = 2$

$s_1(t) = \begin{cases} 1 & 0 \le t \le 1 \\ 0 & \text{otherwise} \end{cases}$ $\quad s_2(t) = s_1(t-1)$

$r_1 \Rightarrow \int_0^{1.5} r(t)\,dt \quad r_2 \Rightarrow \int_1^2 r(t)\,dt$

$r(t) = s_1(t) + n(t)$

$r_1 = \int_0^{1.5} (s_1(t) + n_1(t))\,dt = \int_0^{1} 1\,dt + \int_1^{1.5} 0\,dt + n_1 = 1 + n_1$

$r_2 = \int_1^2 (n(t) + s_1(t))\,dt = n_2$

$r(t) = s_2(t) + n(t)$

$r_1 = \int_0^{1.5} (s_2(t) + n_1(t))\,dt = 0.5 + n_1$

$r_2 = \int_1^2 (s_2(t) + n_2(t))\,dt = 1 + n_2$

$\left.\begin{array}{l} r_1 > r_2 \Rightarrow s_1(t) \\ r_1 < r_2 \Rightarrow s_2(t) \end{array}\right\}$ ML rule

(a) $s_1(t) = \begin{cases} 1 & 0 < t < 1 \\ 0 & \text{otherwise} \end{cases}$

$r(t) = s(t) + As(t-1) + n(t)$

$f_A(a) = \begin{cases} e^{-a} & a > 0 \\ 0 & \text{otherwise} \end{cases}$

$n(t) \sim N_0/2$ delay $= 1$. $s(t) \Rightarrow [0,1]$

$s(t-1) \Rightarrow [1,2]$

$s_1(t)$ sent $\Rightarrow r_1(t) = 1 + n_1$

$r_2(t) = A + n_2$.

$s_2(t)$ sent $\Rightarrow r_1(t) = -1 + n_1$ $r_2(t) = -A + n_2$

$r = \begin{cases} (1, A) + n & , s_1 \\ [-1, -A] + n. & , s_2. \end{cases}$ optimal decision (ML)

$r_1 + Ar_2 > 0 \Rightarrow s_1$

else $\Rightarrow s_2$.

without multipath $\Rightarrow$ BPSK.

$P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$ $E_b = \int_0^1 1^2 dt = 1 \Rightarrow P_e = Q\left(\sqrt{\frac{2}{N_0}}\right)$

with multipath $\Rightarrow (1, A)$. $E = 1 + A^2 \Rightarrow$

$\Rightarrow P_e = Q\left(\sqrt{\frac{2(1+A^2)}{N_0}}\right)$ $A > 0$ $(A \sim e^{-a}) \Rightarrow 1 + A^2 >$

$Q\left(\sqrt{\frac{2(1+A^2)}{N_0}}\right) < Q\left(\sqrt{\frac{2}{N_0}}\right)$ $P_e^{\text{multipath}} < P_e^{\text{no multipath}}$

(6) 8-ary. $E_b = \frac{E_s}{3} \Rightarrow$ Same $E_b \Rightarrow$ Same E_s.

$d_{min}^{(1)} = 2R \sin(\frac{\pi}{8})$ $\quad R = \frac{d_{min}^{(1)}}{2\sin(\frac{\pi}{8})}$

$E_s^{(PSK)} = E_s^{(QAM1)} = E_s^{(QAM2)}$

$E_s^{(PSK)} = R^2 = \frac{(d_{min})^2}{4\sin^2\frac{\pi}{8}}$ $\quad d_{min}^2 = 2\theta$.

E_s^{QAM}

$= a^2+a^2 = 2a^2$ (4 points.)

$3a^2+a^2 = 10a^2$ (4 points)

$6a^2$

$d_{min}^{(2)} = 2a \Rightarrow a = \frac{d_{min}^{(2)}}{2} \Rightarrow E_s = 6 \cdot \frac{(d_{min}^{(2)})^2}{4}$

$= \frac{3}{2} d_{min}^2$.

$d_{min}^{(2)} = \frac{1}{\sqrt{6}} \frac{d_{min}^{(1)}}{\sin(\frac{\pi}{8})}$.

$d_{min}^{(3)} = \sqrt{(3a-a)^2+(0-a)^2} = \sqrt{5}a$.

$E_s = \frac{1}{8}((3a)^2 * 4 + (2a^2) \times 4) = \frac{11}{2}a^2$.

$a = \frac{d_{min}^{(3)}}{\sqrt{5}} \Rightarrow E_s = \frac{11}{2}\left(\frac{d_{min}}{\sqrt{5}}\right)^2$. $E_s = \frac{11}{10}(d_{min}^{(3)})^2$

$\frac{11}{10}(d_{min}^{(3)})^2 = \frac{(d_{min}^{(1)})^2}{4\sin^2\frac{\pi}{8}}$; $d_{min}^{(3)} = \sqrt{\frac{11}{40}} \frac{d_{min}^{(1)}}{\sin\frac{\pi}{8}}$

$P_e \propto Q\left(\frac{d_{min}}{2\sigma}\right)$ → low d_{min}
↳ high P_e

Low BER → large d_{min}

$d_{min}^{(3)} > d_{min}^{(2)} > d_{min}^{(1)}$

QAM 2 have low BER + differs by atmost one bit → gray coding.

PSK & QAM1 → possible.

QAM2 → not possible (few errors by more than 3 pts)

17) $M = 2^N$ take coordinates → Eq

$E_s = Na^2$ $d_{min} = 2a$. $(P_e)_{signal} = N \cdot Q\left(\frac{d_{min}}{2\sigma}\right)$

$\frac{d_{min}}{2\sigma} = \frac{2a}{2\sigma} = \frac{a}{\sigma} \Rightarrow P_e = NQ\left(\frac{a}{\sigma}\right)$

$\sigma^2 = \frac{N_0}{2} \rightarrow \sigma = \sqrt{\frac{N_0}{2}} \rightarrow P_e = NQ\left(\frac{\sqrt{2}a}{\sqrt{N_0}}\right)$

$\rightarrow P_e = N \cdot Q\left(\sqrt{\frac{2E_s}{NN_0}}\right)$ $P_b = \frac{P_s}{\log_2 M} = \frac{P_s}{N} \rightarrow Q\left(\sqrt{\frac{2E_s}{NN_0}}\right)$.